DAFTAR LAMBANG DAN SINGKATAN

SINGKATAN	NAMA
AI	*Artificial Intelligence* (Kecerdasan Buatan)
ANRI	Arsip Nasional Republik Indonesia
API	*Application Programming Interface*
BCE	*Binary Cross-Entropy*
BiGRU	*Bidirectional Gated Recurrent Unit*
BiLSTM	*Bidirectional Long Short-Term Memory*
CBAM	*Convolutional Block Attention Module*
CER	*Character Error Rate* (Tingkat Kesalahan Karakter)
cGAN	*Conditional Generative Adversarial Network*
CLI	*Command-Line Interface*
CNN	*Convolutional Neural Network* (Jaringan Saraf Konvolusional)
CRNN	*Convolutional Recurrent Neural Network*
CTC	*Connectionist Temporal Classification*
CUDA	*Compute Unified Device Architecture*
cuDNN	*CUDA Deep Neural Network library*
CWV	*Core Web Vitals*
DE-GAN	*Document Enhancement GAN*
DIBCO	*Document Image Binarization Contest*
DocEnTr	*Document Enhancement Transformer*
DRD	*Distance Reciprocal Distortion*
DSRM	*Design Science Research Methodology*
ERB	*Enhance to Read Better*
FCN	*Fully Convolutional Network*
FM	*F-measure*
Fps	*pseudo-F-measure*
FR	*Functional Requirement* (Kebutuhan Fungsional)
GAN	*Generative Adversarial Network* (Jaringan Adversarial Generatif)
GPU	*Graphics Processing Unit*
GT	*Ground Truth* (Kebenaran Dasar)
H-DIBCO	*Handwritten Document Image Binarization Contest*
HMM	*Hidden Markov Model*
HTR	*Handwritten Text Recognition* (Pengenalan Teks Tulisan Tangan)

IAM	*IAM Handwriting Database*
ICDAR	*International Conference on Document Analysis and Recognition*
IIIT5K	*IIIT 5K-word dataset*
KHATT	*Arabic Handwritten Text Database*
LSTM	*Long Short-Term Memory*
MAE	*Mean Absolute Error* (Galat Absolut Rata-Rata)
ML	*Machine Learning* (Pembelajaran Mesin)
MLflow	*Machine Learning Flow*
MSE	*Mean Squared Error* (Galat Kuadrat Rata-Rata)
MSFP	*Multi-Scale Feature Pyramid*
NFR	*Non-Functional Requirement* (Kebutuhan Non-Fungsional)
OCR	*Optical Character Recognition* (Pengenalan Karakter Optik)
PALM	*Presentation Attack Liveness Detection in Mobile scenarios*
Pix2Pix	*Image-to-Image Translation with Conditional Adversarial Networks*
PSNR	*Peak Signal-to-Noise Ratio* (Rasio Puncak Sinyal terhadap Derau)
RDB	*Residual Dense Block*
RMSProp	*Root Mean Square Propagation*
RNN	*Recurrent Neural Network* (Jaringan Saraf Berulang)
SDM	Sumber Daya Manusia
SGD	*Stochastic Gradient Descent*
SNR	*Signal-to-Noise Ratio* (Rasio Sinyal terhadap Derau)
SOP	*Standard Operating Procedure* (Prosedur Operasi Standar)
SOTA	*State-of-the-Art*
SSIM	*Structural Similarity Index Measure* (Indeks Kesamaan Struktural)
UNESCO	*United Nations Educational, Scientific and Cultural Organization*
U-Net	*U-shaped Convolutional Network*
VGG	*Visual Geometry Group*
ViT	*Vision Transformer*
VOC	*Visual Object Classes*
WER	*Word Error Rate* (Tingkat Kesalahan Kata)

LAMBANG	**NAMA**
Huruf Yunani (Parameter Model)	
α	Koefisien bobot untuk *loss function*, tingkat signifikansi statistik
β	Koefisien bobot untuk *Binary Cross-Entropy loss*
λ	Koefisien bobot untuk CTC *loss*, parameter regularisasi
θ	Parameter model *neural network*
σ	Deviasi standar
μ	Rata-rata (*mean*)
ε	*Token blank*/kosong dalam CTC, konstanta kecil untuk stabilitas numerik
Δ	Delta (selisih/perubahan nilai)
π	*Alignment path* dalam CTC
π_t	*Token* pada *timestep* t dalam *alignment path*
Operator Matematis	
∇ atau ∂	Operator gradien atau turunan parsial
$\sum$	Operator penjumlahan
$\prod$	Operator perkalian
$\int$	Operator integral
argmax	Argumen yang memaksimalkan fungsi
$\sim$	Terdistribusi menurut
$\rightarrow$	Menuju atau konvergen ke
Fungsi *Loss* dan Kerugian	
$\mathscr{L}$	Fungsi *loss* (kerugian)
$\mathscr{L}_{total}$	Total *loss function*
$\mathscr{L}_{adversarial}$ atau $\mathscr{L}_{adv}$	*Adversarial loss*
$\mathscr{L}_{reconstruction}$	*Reconstruction loss*
$\mathscr{L}_{L1}$	L1 *loss* (*Mean Absolute Error*)
$\mathscr{L}_{L2}$	L2 *loss* (*Mean Squared Error*)
$\mathscr{L}_{CTC}$	CTC (*Connectionist Temporal Classification*) *loss*
$\mathscr{L}_{perceptual}$	*Perceptual loss*
$\mathscr{L}_{pixel}$	*Pixel-wise loss*
$\mathscr{L}_{BCE}$	*Binary Cross-Entropy loss*
$\mathscr{L}_{GAN}$	GAN *loss*
$\mathscr{L}_{cycle}$	*Cycle consistency loss*
Variabel Model dan Komponen GAN	

G	*Generator* dalam GAN
D	*Discriminator* dalam GAN
$G(z)$	*Output* dari *generator* dengan *input* z
$D(x)$	*Output* dari *discriminator* dengan *input* x
$G_{A \to B}$	*Generator* dari domain A ke B
$G_{B \to A}$	*Generator* dari domain B ke A
$V(D,G)$	Fungsi nilai (*value function*) dalam GAN
x	Sampel data asli, *input* citra
y	Label atau *ground truth*, kondisi dalam *conditional GAN*
z	*Noise vector* atau representasi laten

Distribusi Probabilitas

$\mathbb{E}$	Ekspektasi atau nilai harapan
p_{data}	Distribusi data asli
p_z	Distribusi *prior* (*noise*)
$p(y\|x)$	Probabilitas bersyarat *output* y *given input* x
$p_t(\pi_t\|x)$	Probabilitas *token* pada *timestep* t

Fungsi dan Transformasi CTC

$\mathscr{B}$	Fungsi penciutan (*collapse function*) dalam CTC
$\mathscr{B}^{-1}$	Fungsi invers penciutan dalam CTC
T	Jumlah *timestep* atau panjang *sequence*

Metrik Jarak dan Norma

$\|x - G(y)\|_1$	L1 *distance* antara *ground truth* dan *generated image*
$\|x - G(y)\|_2$	L2 *distance* antara *ground truth* dan *generated image*
$N \times N$	Ukuran *patch* dalam PatchGAN

Statistik dan Analisis

d	Cohen's d (*effect size*)
n	Jumlah sampel
p	*p-value* (nilai probabilitas statistik)
r	Koefisien korelasi
R^2	Koefisien determinasi

Notasi Kompleksitas Komputasi

$O(n)$	Notasi Big-O untuk kompleksitas linear
$O(n^2)$	Notasi Big-O untuk kompleksitas kuadratik